

# AP Groundwater Fusion Layer

## Go-Live & Handoff Checklist

**Status:** Prototype, demo-ready (passed external audit). This document lists everything required to take it from prototype to a hosted, auto-refreshing product with real data, and to hand it off to the receiving team.

**[BLOCKER]** blocker for real live data

**[DEPLOY]** needed for a clean hosted  
deploy

**[LATER]** nice-to-have / post-live

## 0 The only true blockers

1. **Official APWRIMS data access** (replaces the short-lived browser-session sample).
  2. **Official mandal boundaries + admin IDs** (replaces public-prototype geometry).
  3. **A host + deploy hook** (or auto-deploy on push).
- Everything else is configuration or polish. Architecture, pipeline, tests, and provenance are already in place.

## 1 Credentials & secrets

Set these in the host's secret store and as GitHub Actions secrets (for the weekly refresh) - never in the repo (the repo is verified clean of secrets).

|                           | Used by                              | Status                                                | Pri.           |
|---------------------------|--------------------------------------|-------------------------------------------------------|----------------|
| APWRIMS_COOKIE            | fetch_apwrims_history.py             | session cookie today; replace with official API/token | <b>BLOCKER</b> |
| EARTHDATA_USER /<br>_PASS | NASA GRACE-DA fetch (.netrc<br>step) | free NASA Earthdata account                           | <b>DEPLOY</b>  |
| DEPLOY_HOOK_URL           | weekly workflow redeploy step        | host deploy webhook                                   | <b>DEPLOY</b>  |
| ANTHROPIC_API_KEY         | ai-brief route                       | optional - degrades gracefully without it             | <b>LATER</b>   |
| ALLOW_INSECURE_TL<br>S    | public-data fetchers                 | leave UNSET in prod; =1 only for local cert<br>issues | <b>LATER</b>   |

## 2 Data sources & auth

| Source            | Provides                       | Auth                                     |
|-------------------|--------------------------------|------------------------------------------|
| APWRIMS (AP-GWD)  | ground-truth groundwater depth | BLOCKER - official access (cookie today) |
| NASA GRACE-DA     | groundwater storage percentile | DEPLOY - Earthdata login (likely)        |
| CHIRPS            | rainfall / recharge            | public                                   |
| TerraClimate      | ET vs rainfall (water balance) | public                                   |
| NASA POWER        | rainfall fallback              | public API                               |
| CGWB / India-WRIS | independent validation network | public portal                            |
| Open-Elevation    | terrain features               | public                                   |

All non-APWRIMS sources are open; confirm each provider's rate limits and attribution/licensing terms before automated production use.

## 3 Replace prototype data with official data

- **[BLOCKER]** Official **APWRIMS groundwater export/API** -> swap the session-sample CSV.
- **[BLOCKER]** Official **APWRIMS / APSAC / RTGS mandal boundaries** -> replace `ap_map_geometry.json` (currently `boundary_source=public_prototype`).
- **[BLOCKER]** Official **mandal admin IDs / codes** -> wire into the join + records.
- **[DEPLOY]** Once official inputs land, review and lift the "prototype / not official" caveats; set `official_flag=true` where justified.

## 4 Auto-refresh pipeline (built - needs hardening)

The weekly GitHub Action (`.github/workflows/phase3_weekly_levels.yml`) runs Mondays: fetch (GRACE-DA / CHIRPS / TerraClimate) -> build features -> predict -> build\_real\_app\_data -> commit app/data + manifest -> redeploy.

- **[BLOCKER]** **APWRIMS cookie is short-lived** - until official API access exists, a human must refresh `APWRIMS_COOKIE` or the step fails.
- **[DEPLOY]** Add the **Earthdata .netrc step** + secrets (placeholder already in the workflow).
- **[DEPLOY]** **Gate publish on CI:** run `pytest + validate_fusion_outputs.py` BEFORE commit/deploy, so a bad fetch can't publish bad data.
- **[LATER]** **Failure alerting** (Slack/email) on a failed weekly run or stale data.

**Cadence reality:** APWRIMS readings are ~monthly, so the meaningful refresh is monthly even though the job runs weekly. The UI shows a "Data as of {month}" stamp.

### Manual pipeline run

```
export APWRIMS_COOKIE='JSESSIONID=...; ...'      # authorized session only
# export EARTHDATA_USER=... EARTHDATA_PASS=...  # if GRACE needs it

python3 phase3_levels/fetch_weekly.py           # fetch + features + predict + publish
python3 phase3_levels/build_real_app_data.py    # (re-publish from existing outputs)

python3 -m pytest -q                             # verify before shipping
python3 scripts/validate_fusion_outputs.py

cd app && npm ci && npm run build
```

## 5 Hosting / deploy

- **[DEPLOY]** Pick a host: **Vercel / Netlify / Render** (auto-deploy on push) or self-host the **standalone** output - run `node .next/standalone/server.js` (NOT `next start`).
- **[DEPLOY]** Custom **domain + HTTPS**.
- **[DEPLOY]** Set `DEPLOY_HOOK_URL` (or rely on host auto-deploy) so the monthly data commit goes live.
- **[LATER]** Decide data delivery: keep commit-data -> redeploy (current) or serve app/data from object storage / an API so fresh data shows without a rebuild.
- **[LATER]** Self-host Google Fonts for fully offline / reproducible builds.

Node 18+; `npm ci` for reproducible installs. Build produces ~660 static pages.

## 6 Security & production hardening

- **[DEPLOY]** Secrets only in host/CI stores (repo is clean; `.gitignore` excludes caches/models/rasters).
- **[LATER]** **AI endpoint:** rate limit is in-memory (per-process) - add a distributed limiter or shared-secret/same-origin check if public at scale.
- **[LATER]** Rotate/destroy the APWRIMS session cookie once official access lands.
- **[LATER]** Add basic error monitoring (e.g. Sentry) and request logging on the host.

## 7 Quality gates already in place (keep running)

- tests/ - 51 tests incl. test\_phase3\_audit.py guardrails: NASA-field == GRACE, measured-vs-NASA separation, secret scanning, env-gated TLS, repo-wide overclaim/attribution scan, boundary coverage, dashboard schema, manifest-hash integrity, CSV provenance banners.
- scripts/validate\_fusion\_outputs.py - output validation.
- app/data/dataset\_manifest.json - sha256 of every generated data file.
- Wire these into CI as the publish gate (see section 4).

## 8 Known limitations to communicate at handoff

- Estimates are modelled gap-fill/forecast for the 639 mandals with APWRIMS history (temporal hold-out, +/-1.3 m); ~20 mandals have no sensor series (shown grey).
- Spatial estimation of sensorless mandals is a research backtest, not deployed.
- GRACE-DA is district-scale context (coarse effective resolution), not mandal depth.
- Pumping vs drought signals are verify-first hypotheses, not attributions.
- Confidence bands are model-derived, not yet calibrated for empirical p10-p90 coverage.

## 9 Deferred engineering (optional, post-live)

- Consolidate the legacy V0 (scripts/) and Phase-3 (phase3\_levels/) pipelines into one documented path (Phase-3 is live, V0 is reference-only).
- Automated responsive + accessibility tests (Playwright + axe); explicit viewport is set, no visual-regression suite yet.
- Recalibrate the model on official APWRIMS data; calibrate confidence bands; validate spatial coverage if statewide sensorless estimates are wanted.
- Phone hamburger drawer (sidebar collapses to an icon rail today; usable but dense on small phones).

## 10 Ownership / runbook (fill in at handoff)

- ☐ Repo owner / maintainer: \_\_\_\_\_
- ☐ Hosting account & who has access: \_\_\_\_\_
- ☐ Secret store location & rotation owner: \_\_\_\_\_
- ☐ Who refreshes APWRIMS\_COOKIE until official access lands: \_\_\_\_\_
- ☐ Monthly-refresh failure escalation contact: \_\_\_\_\_
- ☐ APWRIMS data authorization / MoU owner on AP-WRD side: \_\_\_\_\_